Demonstration of the `refenums` package

This document demonstrates the `refenums` package with the development documents of a reference manager. The content is a simplified version of JabRef's requirements and decision records, written in MADR.

Each section starts with an orange box setting up a kind of item; this produces no output. Each green box shows the LaTeX code producing the output right above it. Items are defined and referenced with shorthands such as `\defReq` and `\refReq`, which `\setupRefEnum{Req}` creates for `\defRefEnum{Req}` and `\refEnum{Req}`.

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1 Features

Features print only their short form (e.g., “Feature 1”) and each becomes a subsection.

```
\setupRefEnum[separator={ }, enclosing=subsection]{Feature}{ONLYSHORT}
% With the default separator "-", this would print "Feature-1"
```

1.1 Feature 1: Save a library

As a researcher, I want to save my library to a `.bib` file, so that I can cite its entries in LaTeX.

```
\defFeature{Save a library}{save}
As a researcher, I want to save my library to a \texttt{.bib} file, so
    that I can cite its entries in \LaTeX{}.
```

1.2 Feature 2: Autosave

As a researcher, I want every change to be saved automatically and a backup to be kept, so that I do not lose work after a crash.

```
\defFeature{Autosave}{autosave}
As a researcher, I want every change to be saved automatically and a
    backup to be kept, so that I do not lose work after a crash.
```

2 Risks

Risks use no separator (e.g., “Risk 1 (R1)”).

```
\setupRefEnum[separator={}]{R}{Risk}
% With the default separator "-", this would print "R-1"
```

Risk 1 (R1): *Concurrent modification.* Another process, e.g., a second JabRef instance on a network share, writes the library file while Feature 1 is running.

```
\defR{Concurrent modification}{concurrent}
Another process, e.g., a second JabRef instance on a network share,
    writes the library file while \refFeature{save} is running.
```

Risk 2 (R2): *Incomplete backup.* A failure while writing a backup during Feature 2 (“Autosave”) replaces a valid backup.

```
\defR{Incomplete backup}{backup}
A failure while writing a backup during \refFeatureFull{autosave}
    replaces a valid backup.
```

3 Requirements

Requirements use the default separator (e.g., “Requirement 1 (Req-1)”).

```
\setupRefEnum{Req}{Requirement}
% Titles are set in italics; titleformat={} would set them upright
```

Requirement 1 (Req-1): *Detect concurrent modification.* If the library file changed since the save started, the save is aborted with an error. Mitigates R1 (“Concurrent modification”).

```
\defReq{Detect concurrent modification}{detect}
If the library file changed since the save started, the save is
    aborted with an error.
Mitigates \refRFull{concurrent}.
```

Requirement 2 (Req-2): *Keep valid backups*. A failure while writing a backup leaves the previous backup untouched. Mitigates R2 (“Incomplete backup”).

```
\defReq{Keep valid backups}{keep}
A failure while writing a backup leaves the previous backup untouched.
Mitigates \refRFull{backup}.
```

4 Decisions

Each decision record becomes a subsection. The options a record considers are defined inline, within the text. Both use the default separator.

```
\setupRefEnum[enclosing=subsection]{ADR}{Decision Record}
\setupRefEnum[inline]{Opt}{Option}
% Without inline, \defOpt would print "Option 1 (Opt-1): ...";
    \defOptInline prints the inline form for a single option
```

4.1 Decision Record 1 (ADR-1): Replace files atomically

Context and Problem Statement How do we write a file so that Req-1 (“Detect concurrent modification”) and Req-2 (“Keep valid backups”) hold?

Considered Options

- *(Opt-1) Overwrite in place*
- *(Opt-2) Write a temporary file and move it over the target*

Decision Outcome Chosen option: Opt-2: "Write a temporary file and move it over the target", because a failed write leaves the original file untouched and the target can be checked for changes right before the move.

```
\defADR{Replace files atomically}{atomic}

\paragraph{Context and Problem Statement}
How do we write a file so that \refReqFull{detect} and
    \refReqFull{keep} hold?

\paragraph{Considered Options}
\begin{itemize}
\item \defOpt{Overwrite in place}{inplace}
\item \defOpt{Write a temporary file and move it over the target}{tmp}
\end{itemize}

\paragraph{Decision Outcome}
Chosen option: \refOptFullColon{tmp}, because a failed write leaves
    the original file untouched and the target can be checked for
    changes right before the move.
```

5 Milestones

Milestones combine print name and short name (e.g., "Milestone M1").

```
\setupRefEnum[combined]{M}{Milestone}
% Without combined, this would print "Milestone 1 (M-1)"
```

Milestone M1: *Safe saving.* Reached when Req-1 and Req-2 are implemented as decided in ADR-1 and covered by unit tests. This mitigates Risks R1 and R2.

```
\defM{Safe saving}{safe}
Reached when \refReq{detect} and \refReq{keep} are implemented as
    decided in \refADR{atomic} and covered by unit tests.
This mitigates \zcref{enum:R:concurrent,enum:R:backup}.
```

6 German requirements

The plural of the print name defaults to the print name with an “s” appended (e.g., “Requirements”). German needs its own plural. `\zcref` follows the language of the text, here set with babel.

```
\setupRefEnum[plural=Anforderungen]{Anf}{Anforderung}
% Without plural, \zcref would print "Anforderungs"
```

Anforderung 1 (Anf-1): *Bibliothek speichern.* Die Bibliothek wird als `.bib`-Datei gespeichert.

```
\selectlanguage{ngerman}
\defAnf{Bibliothek speichern}{speichern}
Die Bibliothek wird als \texttt{.bib}-Datei gespeichert.
```

Anforderung 2 (Anf-2): *Sicherung behalten.* Ein fehlgeschlagenes Schreiben lässt die vorherige Sicherung unverändert.

```
\defAnf{Sicherung behalten}{sicherung}
Ein fehlgeschlagenes Schreiben lässt die vorherige Sicherung
    unverändert.
```

Diese Version erfüllt Anforderungen Anf-1 und Anf-2.

```
Diese Version erfüllt \zcref{enum:Anf:speichern,enum:Anf:sicherung}.
\selectlanguage{english}
```

7 References

The package `refenums` offers several ways to reference an item. They differ in how much of the item they show: its kind, its identifier, its title, or a combination. The following table shows each of them for Req-2 from Section 3.

Aim	Command	Output
Identifier	`\refReq{keep}`	Req-2
Identifier (title)	`\refReqFull{keep}`	Req-2 (“Keep valid backups”)
Identifier: title	`\refReqFullColon{keep}`	Req-2: “Keep valid backups”
Title	`\refReqTitle{keep}`	Keep valid backups
With the kind of the item		
Identifier	`\refReq*{keep}`	Requirement Req-2
Identifier (title)	`\refReqFull*{keep}`	Requirement Req-2 (“Keep valid backups”)
Identifier: title	`\refReqFullColon*{keep}`	Requirement Req-2: “Keep valid backups”

Each shorthand has a long form, which takes the kind of the item as first argument. For instance, `\refReqFull{keep}` is short for `\refEnumFull{Req}{keep}`. The command `\refReq*{keep}` is also short for `\zcref{enum:Req:keep}` of zref-clever, which also references several items at once, as in Section 5.

The following list references every item of this document in the form “Identifier (title)”.

- Features: Feature 1 (“Save a library”), Feature 2 (“Autosave”)
- Risks: R1 (“Concurrent modification”), R2 (“Incomplete backup”)
- Requirements: Req-1 (“Detect concurrent modification”), Req-2 (“Keep valid backups”)
- Decision records: ADR-1 (“Replace files atomically”)
- Options: Opt-1 (“Overwrite in place”), Opt-2 (“Write a temporary file and move it over the target”)
- Milestones: M1 (“Safe saving”)

```
\begin{itemize}
\item Features: \refFeatureFull{save}, \refFeatureFull{autosave}
\item Risks: \refRFull{concurrent}, \refRFull{backup}
\item Requirements: \refReqFull{detect}, \refReqFull{keep}
\item Decision records: \refADRFull{atomic}
\item Options: \refOptFull{inplace}, \refOptFull{tmp}
\item Milestones: \refMFull{safe}
\end{itemize}
```